Here's a complete breakdown of the **difference between Azure App Service and Azure Static Web Apps**, with **examples**, **use cases**, and **required files** for deployment.

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1. Key Differences Between Azure App Service and Azure Static Web App

Feature	Azure App Service	Azure Static Web App
Purpose	Host full-stack dynamic web apps, REST APIs, mobile backends	Host static frontend apps with optional serverless APIs
Content Type	HTML, CSS, JS, server-side code (.NET, Node.js, Java, Python)	Static content (HTML, CSS, JS, images) + optional Azure Functions
Backend	Built-in support for dynamic server-side backends	Supports backend via Azure Functions (API routes)
Build & Deployment	Manual deployment (FTP, Git, CI/CD) or Azure DevOps/GitHub Actions	GitHub Actions CI/CD pipeline automatically configured
Custom Domains & SSL	Supported	Supported (free SSL and custom domains included)
Pricing	Paid (more flexible and powerful)	Free tier available, optimized for frontend apps
Authentication	Supports built-in auth (Azure AD, Facebook, Google, etc.)	Built-in auth with simple config in routes.json
Best Use Case	Full-stack apps, .NET, Java apps, RESTful APIs	Static websites, SPAs (React, Angular, Vue), JAMstack apps

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2. Use Case Examples

A. Azure App Service Example

- Hosting a **Spring Boot REST API** or **Node.js Express server**
- Hosting a **PHP or .NET web application**

B. Azure Static Web App Example

- Hosting a **static gym website** made in HTML/CSS/JavaScript
- Hosting a **React/Angular** frontend with backend APIs in **Azure Functions**

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3. Required Files and Directory Structures

For Azure App Service

You deploy your full app via FTP, ZIP deploy, or CI/CD pipeline.

Directory Structure Example (Static Gym Website):

```
static-gym-html-css-app/
├── aboutus.html
├── index.html
├── contact.html
├── abc.css
├── images/
│   └── logo.png
└── web.config (if needed for IIS hosting or redirects)
```

Required Files:

- No mandatory config files, but for CI/CD you may use:
  - azure-pipelines.yml for Azure DevOps
  - .github/workflows/deploy.yml for GitHub Actions

For Azure Static Web Apps

Directory Structure Example:

```
static-gym-html-css-app/
├── index.html
├── aboutus.html
├── abc.css
├── assets/
│   └── gym.jpg
├── api/                ← (Optional) Azure Functions API
│   └── hello-world/index.js
├── staticwebapp.config.json  ← For routes, auth, etc.
├── .github/
│   └── workflows/
│       └── azure-static-web-apps.yml
```

Required Files:

- index.html (entry point)
- staticwebapp.config.json (for routing, headers, auth config)
- .github/workflows/azure-static-web-apps.yml (auto-generated GitHub Actions file)

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4. Example Files

staticwebapp.config.json (for Static Web App)

```
{
  "navigationFallback": {
    "rewrite": "/index.html"
  },
  "routes": [
    {
      "route": "/admin",
      "allowedRoles": ["admin"]
    }
  ],
  "responseOverrides": {
    "404": {
      "rewrite": "/404.html"
    }
  }
}
```

GitHub Actions Workflow for Static Web App

.github/workflows/azure-static-web-apps.yml

```
name: Azure Static Web Apps CI/CD

on:
  push:
    branches:
      - main

jobs:
  build_and_deploy:
    runs-on: ubuntu-latest
    name: Build and Deploy Job
    steps:
      - uses: actions/checkout@v3
      - name: Build And Deploy
        uses: Azure/static-web-apps-deploy@v1
        with:
          azure_static_web_apps_api_token: ${{ secrets.AZURE_STATIC_WEB_APPS_API_TOKEN }}
          repo_token: ${{ secrets.GITHUB_TOKEN }}
          action: "upload"
          app_location: "/"
          api_location: "api"
```

```
    output_location: "/"
```

Summary: When to Use What?

Requirement	Use
Static frontend with API integration	Azure Static Web Apps
Full-stack dynamic apps (backend + frontend)	Azure App Service
Hosting HTML/CSS/JS site quickly with SSL and CI/CD	Azure Static Web Apps
Hosting .NET, PHP, Node.js with backend logic	Azure App Service

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Would you like a complete working demo ZIP or GitHub repo layout for either service?